

Visual Attention is Not Memoryless: Higher-Order Sequence Modeling of TeamLead Gaze in VR Cardiac Arrest Simulation

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Abstract. In multiperson virtual-reality cardiac arrest simulations, team leads support team situational awareness by continuously shifting visual attention among the patient, vital-sign monitors, resuscitation equipment, and teammates. We study these transitions by mapping each fixation to one of seven clinically defined areas of interest (AOIs), such as the patient, vitals monitor, or resuscitation equipment, and modeling sequences of AOI changes. First-order Markov models, commonly used in gaze analysis, treat this process as memoryless: the next AOI is assumed to depend only on the current AOI. This assumption may miss short supervisory routines, such as briefly checking the monitor before returning to the patient or another resuscitation-relevant target. We analyze TeamLead AOI sequences from $n=14$ simulation sessions (one TeamLead sequence per session), using k -th-order Markov models with Laplace smoothing and leave-one-session-out cross-validation. Our results show that, among candidate orders $k \in \{1, 2, 3, 4, 5\}$, a second-order model ($k=2$) maximizes held-out mean log-likelihood and outperforms the first-order baseline on all three metrics (mean Δ log-likelihood +0.086, accuracy +0.11, macro- F_1 +0.14; all $p < 0.001$). A stage-conditioned extension, which estimates separate transition counts for each annotated stage of the cardiac-arrest scenario, also beats the first-order baseline but is strictly dominated by the AOI-only second-order model, suggesting redundancy between explicit stage splits and compact AOI history at this sample size. Higher-order gains concentrate in short return-to-anchor triplets (e.g., monitor, patient, monitor), consistent with cyclic supervisory monitoring rather than memoryless transitions.

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Keywords: higher-order Markov model · gaze analysis · situational awareness · virtual reality simulation · cardiac arrest training

1 Introduction

In high-stakes, time-critical clinical environments, situational awareness (SA) is essential for effective team performance [12]. Cardiac arrest is a prototypical example of such an environment, where patient status can deteriorate within minutes, and team members must rapidly recognize rhythm changes, coordinate role-specific actions, and execute time-critical interventions [10]. These demands are not distributed evenly across team members; rather, they place a distinctive burden on the TeamLead, who must maintain a global view of the evolving scenario while coordinating the actions of their teammates. Therefore, understanding how TeamLeads allocate and update attention is central to explaining how SA is maintained during resuscitation.

In this role, the TeamLead must monitor task progress, coordinate team activity [8], and reallocate attention as clinical demands evolve [5]. Because visual attention provides an observable trace of Level 1 SA (perception of environmental cues) [4], gaze data have become a valuable proxy for studying how SA is maintained during team-based clinical work [1, 19]. However, SA in resuscitation is not static. TeamLeads must continuously update their understanding of patient status, team actions, and scenario progress as events unfold over time. This dynamic nature makes the temporal organization of gaze especially important, because where the TeamLead looks next may depend not only on the current clinical cue but also on the recent sequence of attended information.

To examine whether TeamLead gaze exhibits such history-dependent temporal structure, we apply higher-order network (HON) analysis to TeamLead AOI sequences in multiperson virtual reality (VR) cardiac arrest simulation. HON extends first-order transition modeling by representing cases in which the next state depends on recent path history in addition to the current state [20], making it well suited for detecting short-range dependencies such as recurring routines between AOIs. We situate the analysis in VR-based simulation, where SA and teamwork are central to performance and where immersive, repeatable scenarios enable fine-grained behavioral data collection [13]. We compare a first-order Markov baseline with HON under leave-one-session-out cross-validation, and test whether adding explicit annotated clinical-stage context improves prediction. Specifically, we ask two research questions:

- RQ1.** How does higher-order modeling improve next-AOI prediction beyond a first-order baseline in TeamLead AOI sequences?
- RQ2.** How does adding clinical-stage context affect the predictive value of higher-order AOI sequence modeling?

We frame this study as an initial, single-scenario demonstration rather than broad evidence about clinician gaze in emergencies: the analysis draws on 14 TeamLead sessions from one VR-based STEMI protocol, and we position compact higher-order gaze modeling as a proof of concept to be tested more widely.

2 Related Work

Gaze and situational awareness in clinical simulation. Endsley’s three-level model frames SA as the perception of cues, the comprehension of their meaning, and the projection of future state [4]. Because gaze provides an observable trace of Level 1 perception, eye-tracking has become a widely used proxy for SA in time-critical clinical settings [1, 19], where studies link aggregate gaze metrics (fixation counts, dwell times, AOI proportions) to leader behaviors and team performance, and multimodal extensions combine gaze with pose or speech to characterize teamwork during training [13, 12]. These studies typically describe *where* attention is allocated, but do not directly test the sequential generative process underlying those allocations.

Transition- and network-based analyses of gaze. A second body of work models visual attention as transitions between AOIs. Gaze transition entropy quantifies how unpredictable next-fixation behavior is given the current AOI [9], and Transition Network Analysis (TNA) has been proposed as a general framework for modeling transition structure in learning analytics data [15]; prior work applied first-order TNA to multi-role gaze in a multiperson VR resuscitation dataset to characterize role- and stage-specific differences in attentional structure [5], and related ordered-network approaches extend the idea to team communication [12]. A common commitment unites these methods: transitions are estimated from *first-order* statistics, treating the next AOI as dependent on the current AOI alone. Such quantities cannot, by construction, distinguish two visits to the same AOI that originate from different short histories. We directly test whether that assumption is empirically warranted for TeamLead gaze, using held-out predictive performance.

Higher-order models for path-dependent behavior. Path-dependent processes that violate the first-order assumption have been studied extensively. Xu et al. [20] introduce higher-order networks (HON) as a representation that selectively retains longer-history contexts when those contexts carry additional predictive power, and Scholtes [16] formalizes the choice of order k as a multi-order model-selection problem. Two observations motivate our choices. First, higher-order gains typically concentrate in a few “surprising” contexts where future state genuinely depends on more than the current node [20, 16], a pattern well suited to supervisory monitoring routines. Second, with a small alphabet (seven AOIs) and a modest corpus, fixed-order Markov models with smoothing are a simple, interpretable instantiation of higher-order modeling. Our contribution is not a new HON algorithm, but an empirical test of whether even this minimal extension yields predictive benefits on TeamLead gaze beyond first-order analyses, and a characterization of *which* short histories drive that benefit.

3 Methods

Data and Areas of Interest. Eye-tracking data came from multiperson VR cardiac arrest simulations in which teams of four trainees managed scenarios in standardized roles (TeamLead, Airway, CPR, Defib) following American

Table 1. Areas of Interest (AOIs) used for gaze-sequence analysis.

AOI	Description
Equipment - Airway	Airway management equipment (e.g., bag-valve masks).
Equipment - CPR	Hands and UI (user interface) elements directly involved in CPR.
Equipment - Defib	Defibrillator unit and associated controls.
Equipment - Meds & IV	Medication- and IV-related objects.
Patient Vitals Monitor	ECG, SpO2, respiration, and related vitals.
Other Team Members	3D avatars of other people in session.
Patient	Virtual patient body.

Heart Association guidelines [11]. During resuscitation, Airway, CPR, and Defib providers focus on role-specific interventions (e.g., ventilation, compressions, and defibrillation), whereas the TeamLead coordinates the team, monitors patient status and overall task progress, and directs time-critical treatment decisions [10, 8]; we therefore analyze TeamLead gaze as the supervisory attention stream most likely to integrate information across the patient, monitor, equipment, and teammates. Gaze was recorded with HP Reverb G2 Omnicept headsets using the Cognitive3D platform [3]; the AOI scheme and preprocessing pipeline follow those documented for this dataset [5]. The scenario follows a STEMI cardiac arrest [18] that progresses from ventricular tachycardia with a weak pulse through pulseless VT, asystole, and ventricular fibrillation before return of circulation. Data collection was approved by the Institutional Review Board (IRB protocol HUM00193383). We analyze the 14 sessions that provided complete, continuous TeamLead eye-tracking coverage together with clinical-stage annotation, one TeamLead sequence per session; recordings with interrupted tracking or missing stage annotations were excluded. Sessions averaged 13.1 min ($SD = 3.6$). The TeamLead role produced 31,157 raw fixations; after discarding fixations without a valid AOI and collapsing consecutive fixations on the same AOI, 8,798 AOI dwell events and 8,784 AOI transitions remained across the 14 sessions and served as the input to all sequence models.

Gaze was mapped to seven AOIs (Table 1) serving as perceptual anchors: airway, CPR, defibrillation, and medication/IV equipment; the patient vitals monitor; the patient; and other team members.

AOI Sequences and Higher-Order Markov Modeling. To examine whether TeamLead gaze follows a memoryless process or a history-dependent sequence process, we modeled each TeamLead’s gaze behavior as a time-ordered sequence of seven clinically meaningful AOIs. This framing treats each fixation as an observation of where the TeamLead allocated visual attention and asks whether the next AOI can be predicted from the immediately preceding AOI alone, as assumed by a first-order Markov model, or whether prediction improves when recent AOI history is included.

Each valid fixation was mapped to a single AOI label and ordered by timestamp to form a role-specific AOI sequence; fixations without valid labels and saccade samples were excluded. Consecutive repeated labels were collapsed so

that each step corresponds to a change of AOI (a transition event), and fixations landing on a non-clinical region outside the seven AOIs (an artifact of VR locomotion, e.g., teleport-based navigation) were removed before collapsing. Analyses focus on the **TeamLead** sequence for each session.

Let $(x_1, \dots, x_T)$ denote a TeamLead AOI sequence over the seven labels. For a fixed context length $k \geq 1$, each prediction position $t \in \{k+1, \dots, T\}$ uses context $\mathbf{c}_t = (x_{t-k}, \dots, x_{t-1})$ and outcome x_t . Counts $N(\mathbf{c}, a)$ accumulate how often label a follows context $\mathbf{c}$ in training. Because higher-order contexts can be sparse, especially when evaluating held-out sessions, transition probabilities were estimated with additive (Laplace) smoothing [2] using $\alpha = 0.5$ over the closed vocabulary of seven AOIs. This assigns a small amount of probability mass to unseen but possible transitions and prevents zero probabilities from dominating the held-out log-likelihood:

$$\hat{P}(x_t = a \mid \mathbf{c}_t) = \frac{N(\mathbf{c}_t, a) + \alpha}{\sum_{a'} N(\mathbf{c}_t, a') + 7\alpha}.$$

When $k = 1$, the model reduces to a standard first-order Markov chain in which the next AOI depends only on the current AOI. When $k > 1$, each distinct length- k history defines a higher-order context, allowing the predicted next AOI to depend on recent sequence history. Throughout, we refer to this smoothed fixed-order model as our higher-order network (HON) instantiation: it captures history dependence through explicit length- k contexts rather than the pruned, rule-based HON graph of Xu et al. [20]. The *predicted* next AOI is the argmax under $\hat{P}(\cdot \mid \mathbf{c}_t)$. Session-level **mean log-likelihood** averages $\log \hat{P}(x_t \mid \mathbf{c}_t)$ over scored transitions; **accuracy** is the fraction of correct argmax predictions; **macro-F₁** averages per-class F₁ over the seven AOIs. We report these three metrics because they are complementary: mean log-likelihood is a proper scoring rule [6] sensitive to the full predicted distribution and serves as our model-selection criterion; accuracy summarizes hard next-AOI prediction but is dominated by frequent AOIs; and macro-F₁ weights all seven AOIs equally, revealing whether history helps recover less frequent destinations.

Evaluation and order selection. Models were fit under **leave-one-session-out (LOSO)** cross-validation: for each of 14 sessions, parameters were estimated from all other sessions and evaluated on the held-out TeamLead sequence, avoiding within-session leakage. Order k was selected from $\{1, 2, 3, 4, 5\}$ by maximizing the mean held-out log-likelihood across folds. Bayesian Information Criterion (BIC) [17] summaries on training folds were retained for reference but were not the primary rule, as BIC favored $k=1$ despite stronger held-out likelihood for larger k .

To test whether clinical stage improves prediction beyond AOI history, we fit a stage-conditioned model. Clinical stages were annotated from the scenario’s scripted rhythm timeline, which assigns each rhythm phase (e.g., ventricular tachycardia, asystole, ventricular fibrillation, and return of circulation) start and end timestamps; because different rhythms prioritize different information (e.g., defibrillator and monitor during shockable rhythms versus patient and airway

during compressions), stage could in principle sharpen next-AOI prediction. The model estimates separate k -th-order count tables for each stage while sharing the same order k . At each prediction position t , the model conditions on the clinical stage annotated at that time index. Sparse (stage, $\mathbf{c}_t$) combinations use hierarchical back-off: stage-specific counts when available, otherwise the global k -th-order model, then the global first-order model, then a uniform distribution.

For each session and metric, we computed paired differences between the higher-order model at the selected k and the first-order baseline ($k=1$) on the same held-out sequence. Because sessions are not assumed normal, we tested whether median paired differences differed from zero using the **Wilcoxon signed-rank** test on the 14 session-level deltas.

We also summarized recurring higher-order motifs induced by the fitted model. For each observed length- k context with sufficient support, we recorded the most probable next AOI and its empirical support. When $k=2$, such contexts correspond to weighted AOI triplets $x_{t-2} \rightarrow x_{t-1} \rightarrow x_t$ that highlight short return-to-anchor routines invisible to a memoryless model; they are used descriptively in Section 4 to interpret *where* the model gains information.

4 Results & Discussion

We compare two higher-order specifications, both using the selected order $k=2$: an *AOI-only* model that conditions on AOI history alone, and a *stage-conditioned* model that additionally splits transition counts by clinical stage. We organize the results around the two research questions: Section 4.1 establishes the optimal order; Section 4.2 addresses RQ1 with the AOI-only model; Section 4.3 addresses RQ2 by adding stage conditioning; Section 4.4 interprets which contexts drive the gain.

4.1 Order selection: $k=2$ is optimal

Held-out mean log-likelihood as a function of k (Fig. 1) is non-monotone: a sharp gain from $k=1$ (≈ -1.017) to $k=2$ (≈ -0.930), followed by steady decline for $k=3, 4, 5$ as the number of distinct length- k contexts grows and per-context counts become sparse on a 7-AOI alphabet with 14 sessions. The relevant memory in TeamLead AOI sequences is therefore short: most predictable structure is captured by conditioning on the previous *two* fixations. We use $k=2$ for all subsequent comparisons, consistent with HON practice favoring compact, well-supported contexts [20].

4.2 Higher-order modeling improves next-AOI prediction (RQ1)

In the AOI-only model, HON($k=2$) outperforms the first-order baseline on all three metrics (Fig. 2; Table 2). Mean log-likelihood improves by +0.0858 nats (natural-log units) per scored transition ($p = 0.00037$), a multiplicative likelihood gain of roughly 9% per step. Argmax accuracy improves from a median

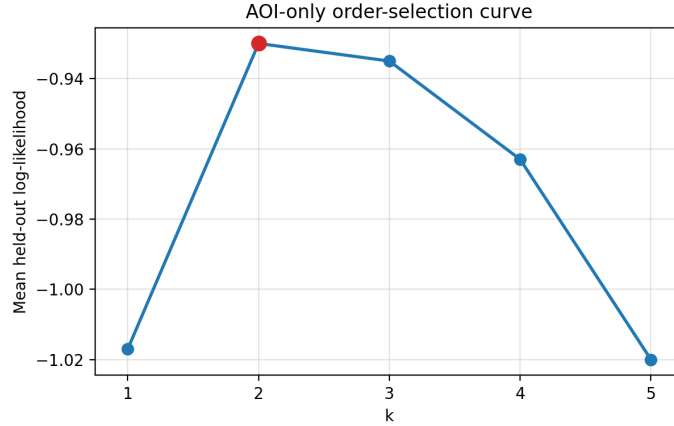


Fig. 1. Order-selection curve for the AOI-only model: mean held-out log-likelihood across LOSO folds vs. Markov order k . Performance peaks at $k=2$ and degrades for $k \geq 3$ as context tables fragment.

AOI-only: HON($k=2$) vs first-order across 14 sessions

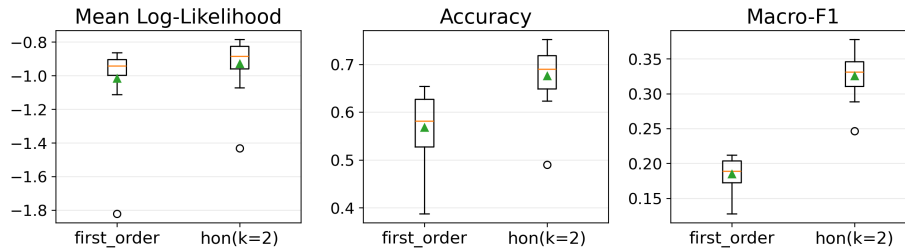
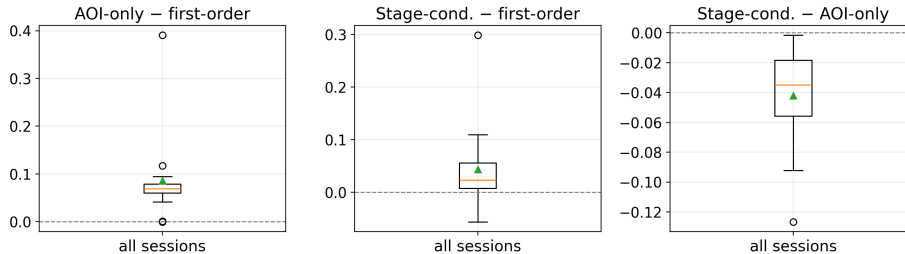


Fig. 2. AOI-only HON($k=2$) versus first-order baseline across 14 LOSO folds on three session-level metrics. Boxes show medians and IQR; triangles mark means. HON improves all three metrics on the majority of sessions.

around 0.58 to 0.69 ($\Delta = +0.108$, $p = 0.00012$), and macro- F_1 —which weights minority AOIs equally—increases by roughly three-quarters, from ≈ 0.19 to ≈ 0.33 ($\Delta = +0.141$, $p = 0.00012$). The macro- F_1 gain is informative: HON does not merely sharpen predictions for the most frequent AOIs (e.g., *Patient Vitals Monitor*) but recovers structure around less frequent destinations whose distribution depends on which AOI the TeamLead just came from. This answers RQ1: the first-order assumption underlying transition-based gaze analyses [9, 15, 5] is empirically too restrictive for TeamLead gaze, and a compact $k=2$ history yields a reliable, substantial improvement. Bootstrap 95% confidence intervals over the 14 folds exclude zero for all three metrics (log-likelihood $[+0.05, +0.14]$, accuracy $[+0.09, +0.13]$, macro- F_1 $[+0.13, +0.15]$), so the gain does not rest on a few sessions. Practically, correctly anticipating these less frequent, history-dependent transitions—not only the dominant monitor fixation—is what an attention-aware

Table 2. AOI-only model: paired session-level deltas ($\text{HON}(k=2) - \text{first-order}$) across 14 LOSO folds, two-sided Wilcoxon signed-rank tests.

Metric	Mean Δ	Wilcoxon p	Direction
Mean log-likelihood	+0.0858	0.00037	HON > first-order
Accuracy	+0.1081	0.00012	HON > first-order
Macro- F_1	+0.1411	0.00012	HON > first-order

**Fig. 3.** Per-session log-likelihood deltas. **Left:** AOI-only $\text{HON}(k=2) - \text{first-order}$, mean +0.0858 ($p=0.00037$). **Middle:** stage-conditioned $- \text{first-order}$, mean +0.0437 ($p=0.0166$). **Right:** context effect, stage-conditioned $- \text{AOI-only}$, mean -0.0421 ($p=0.00012$). Stage-conditioning beats first-order but is dominated by the AOI-only model.

debriefing or feedback tool would require to flag atypical monitoring, which motivates the clinical directions discussed below.

4.3 Adding clinical-stage context does not help (RQ2)

The stage-conditioned model also outperforms the first-order baseline on log-likelihood (mean $\Delta = +0.0437$, $p = 0.0166$; Fig. 3, middle), but is consistently *worse* than the AOI-only model on the same held-out sessions: the log-likelihood difference (stage-conditioned $- \text{AOI-only}$) is centered below zero on all but one fold (mean -0.0421 , $p = 0.00012$; right). Thus, while clinical stage carries some signal relative to a memoryless model, conditioning on stage *costs more in count fragmentation than it adds* at the current sample size and stage granularity. Two non-exclusive mechanisms fit this pattern. (i) *Sparsity dominates*: stage-specific length-2 tables contain many low-support cells that smoothing pushes toward less informative pooled estimates. Consistent with this, the AOI-only model almost never invokes its uniform fallback (an unseen length-2 context on only $\approx 0.2\%$ of held-out steps, at most 2.2%), and the stage-conditioned model still draws on stage-specific counts for 97% of steps rather than backing off—so its shortfall reflects noisier within-stage estimates, not collapse to weaker fallback models. (ii) *Stage information is partly redundant with AOI memory*: if the stage signal is largely realized through which AOI pairs become locally common (e.g., increased Defib-Monitor coupling during ventricular fibrillation), a $k=2$ AOI-only model already absorbs it. The negative net effect therefore identifies

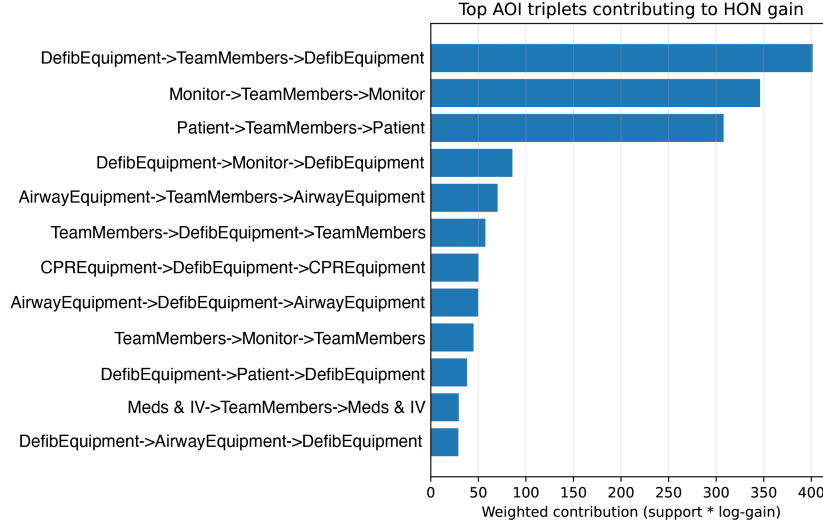


Fig. 4. Top higher-order motifs (length-2 contexts) ranked by weighted contribution to the held-out log-likelihood gain of HON($k=2$) over first-order. Most are short return-to-anchor triplets ($A \rightarrow B \rightarrow A$) involving *Equipment* – *Defib*, *Patient Vitals Monitor*, and *Patient*, indicating structured monitoring loops.

a *representational redundancy* rather than falsifying the relevance of stage, and suggests future encodings via change-point segmentation or soft mixtures rather than hard partitions.

4.4 Mechanism: short recurring routines drive the gain

To localize *where* the AOI-only gain originates, we ranked length-2 contexts by their weighted contribution to the LOSO log-likelihood difference between HON($k=2$) and first-order (Fig. 4). The dominant contributors share a *return-to-anchor* shape $A \rightarrow B \rightarrow A$, in which the TeamLead briefly looks away from a focal AOI (A) to consult an adjacent AOI (B) and then returns. The strongest routines involve *Equipment* – *Defib*, *Patient Vitals Monitor*, and *Patient* as anchors, with monitor and patient typically serving as B . A first-order model cannot represent these triplets: from state B it must commit to a single outgoing distribution regardless of which AOI was visited two steps ago. The $k=2$ model resolves exactly this ambiguity, which explains why the macro- F_1 gain exceeds the accuracy gain: return-to-anchor structure reroutes probability mass toward the historically correct A and away from the marginal-mode prediction.

Behaviorally, these motifs are consistent with a leader’s monitoring rhythm: brief consultative glances away from a primary referent (the defibrillator during a shock cycle, or the patient during a pulse check) followed by an active return to verify the consequence of an action, consistent with supervisory monitoring as cyclic verification rather than open-ended exploration [14, 10]. Methodologically,

this clarifies the scope of first-order TNA: descriptive role and stage contrasts remain valid as summaries, but the *generative* mechanism producing them operates over short windows of AOI history rather than single transitions.

4.5 Limitations & Future Work

Three limitations apply. First, the corpus is restricted to the TeamLead role and a single VR-based STEMI scenario, so generalization to other roles, scenarios, and real settings remains to be tested [7]. Second, with seven AOIs and 14 sessions, $k=3$ contexts are systematically under-supported, constraining the order search to short histories; richer corpora may permit deeper memory. Third, the stage-conditioned variant uses fixed annotations; we do not yet test data-driven change-point or soft-mixture alternatives. Finally, the smoothing strength ($\alpha=0.5$) was fixed a priori rather than tuned, and a sensitivity analysis is left to future work. We also selected the order k and evaluated it on the same LOSO folds; the effect is large and its bootstrap intervals are well clear of zero, but fully nested cross-validation would remove any residual model-selection optimism.

Building on these findings, a key next step is to establish the clinical value of the identified return-to-anchor ($A \rightarrow B \rightarrow A$) motifs with domain experts, for example by having clinician instructors judge whether these monitoring routines distinguish effective from ineffective situational awareness or expert from novice team leadership in simulation. Confirming that link would enable translation into practice: debriefing visualizations that surface a leader’s monitoring rhythm, attention-based training feedback, and, ultimately, real-time indicators of degraded situational awareness during resuscitation. Methodologically, the approach should also extend beyond the TeamLead role and the single STEMI scenario, incorporate deeper histories as larger corpora permit, compare soft or change-point stage encodings against hard partitions, and integrate gaze with communication and action streams for joint models of team coordination.

5 Conclusion

We showed that TeamLead gaze in VR cardiac arrest simulation carries short-range sequential structure beyond a first-order Markov model: a compact second-order model improves next-AOI prediction under LOSO evaluation, and the gain localizes to recurring monitoring motifs. Explicit clinical-stage conditioning did not add held-out predictive value here, likely reflecting count sparsity and overlap between stage effects and AOI-context patterns already captured at $k=2$.

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